

cse 242 hw2

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1. (a)

$$\begin{aligned} J(w) &= \|Xw - y\|_2^2 + \lambda \|w\|_2^2 \\ J(w) &= (Xw - y)^T (Xw - y) + \lambda w^T w \\ J(w) &= w^T X^T X w - w^T X^T y - y^T X w - y^T y + \lambda w^T w \quad (1) \\ J(w) &= w^T X^T X w - 2y^T X w - y^T y + \lambda w^T w \\ \frac{\partial J(w)}{\partial w} &= 2X^T X w - 2y^T X + 2\lambda w = 0 \end{aligned}$$

then we see

$$\begin{aligned} 2X^T X w + 2\lambda w &= 2y^T X \\ (X^T X + \lambda I)w &= y^T X \end{aligned} \quad (2)$$

then we get the solution in the hw.

- (b) as $\lambda \rightarrow 0$, $w_{ridge} \rightarrow (X^T X)^{-1} X^T y$. as $\lambda \rightarrow \infty$, $w_{ridge} \rightarrow 0$.
the former is when penalty goes to zero and the latter is when error dominates.

(c)